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AET-5010-01
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27 April 2023

Audio electronic device measurement project

Device Under Test (DUT):

The DUT used to perform the analog audio device measurement with Audio Precision (AP) was a Fusion all-analogue stereo outboard processor by Solid State Logic (SSL). This device can be used in conjunction with an audio interface to add colour, saturation, depth, compression, and distortion to expand the creative possibilities of sound processing in the recording studio [1]. SSL is an established brand in the music industry, known for creating products that are widely used by sound engineers in recording studios.

Specifications: Audio Performance (from manufacturer)

Default test conditions:
Source impedance of Test Set: 50Ω
Input impedance of Test Set: 100kΩ
Reference frequency: 1kHz
Reference level: 0dBu
All unweighted measurements are specified as 22Hz to 22kHz band limited RMS and are expressed in units of dBu.
The onset of clipping (for headroom measurements) should be taken as 1% THD.
Measurements taken with Input and Output trim at center position.
All levels are intended balanced.

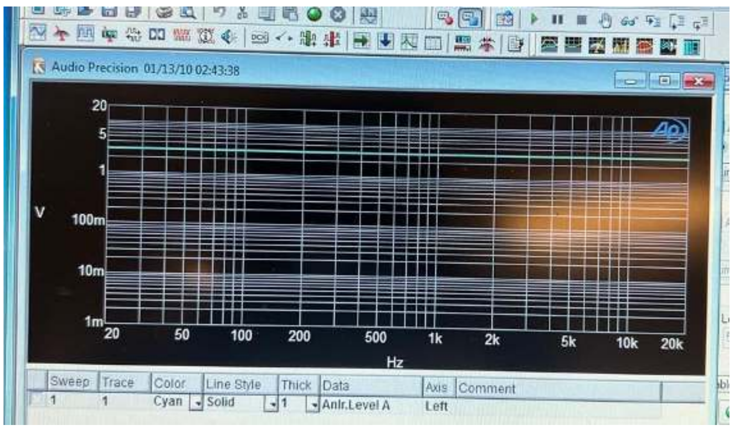
Measurement	Conditions	Value
Input Impedance		10kΩ
Output Impedance		75Ω
Max Input Level	1% THD	27.25 dBu
Max Output Level	1% THD	27.25 dBu
Frequency Response	All circuits off - 20Hz to 20kHz - -3dB low rolloff - -3dB high rolloff	 - ±0.05dB - < 5Hz - > 180kHz
THD+Noise	All circuits off - +20dBu, 1kHz (filter 22Hz to 22kHz) Bypass - +20dBu, 1kHz (filter 22Hz to 22kHz) Vintage Drive - Indicator Off - Indicator Green - Indicator Orange - Indicator Red	 - < 0.005%, 0.0025% typical - 0.0005% typical - < 0.2% typical - 0.2% to 0.5% typical - 0.5% to 2% typical - > 2% typical
CMRR	- 20Hz - 1kHz	- > 78dB, 85dB typical - > 78dB, 88dB typical
Noise	Analyser filters/BW: 22Hz to 20kHz (AES17). - All circuits off - Bypass - Vintage Drive, mid Drive, mid Density	 - < -86dBu, -91dBu typical - -94dBu typical - -75dBu typical

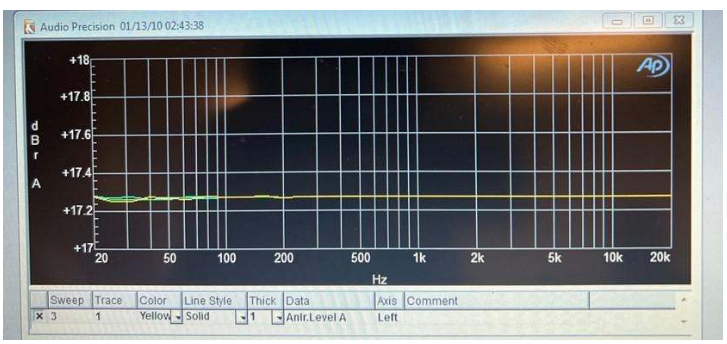
Measurement	Conditions	Value
Crosstalk	Analyser filters/BW: 22Hz to 20kHz (AES17). Generator +24dBu into one channel, analyzer on the other channel (generator connected but not active). All circuits off / bypass - 1 kHz - 20Hz	 - < -110dB, -115dB typical - -110dB typical
Stereo Matching	- All circuits off / Bypass - Vintage Drive on	- < 0.01dB - < 0.25dB typical

Source adapted from: [1]

Audio Measurements

Frequency Response:

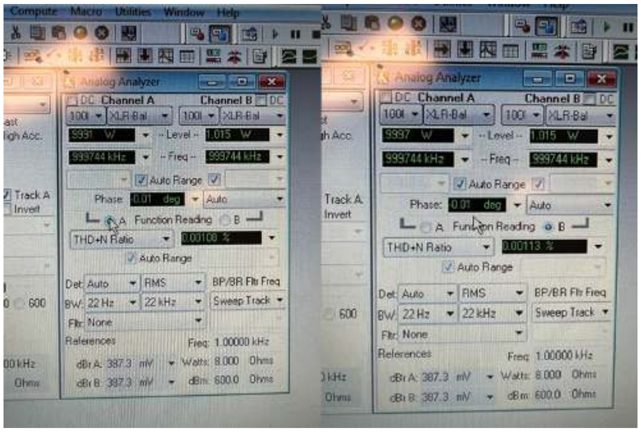




The graph above shows the frequency response of the DUT. First, we adjusted the input level that produces a given output level to about 1 W (unity gain) to match the generator, then adjusting the DUT to about 1% THD+N. To setup the frequency response sweep in AP, data 1 was set to analyzer, level A and source 1 to analog generator. Frequency. The generator sent a 1 kHz sine wave at a level of 1 Vrms. The DUT volume was increased to about 1 W. This gave us a flat response with minimal variations. Overall, the sweep showed some similarity with the manufacturers, however our response was about +17 dB. This could be due to the manufacturer's audio performance default test being measured at different metrics.

+17.3, +17.2 dB, 20 Hz to 20 kHz, at 2 dB below rated power. (from lower graph)

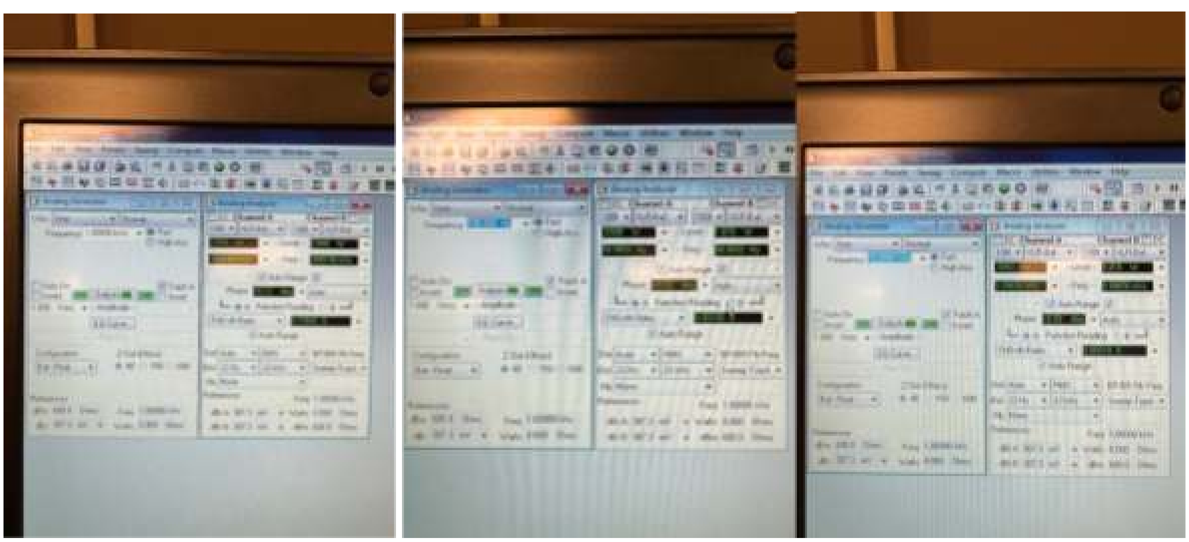
THD+N:



The charts above show the total harmonic distortion plus noise for channels A and B. To setup this measurement the function reading was changed to THD+N Radio. With a bandwidth of 22 Hz to 22 kHz, then we observed the readings. Results of this measurement were about 0.001% for both channels, which is lower than the manufacturers due to the parameters being different.

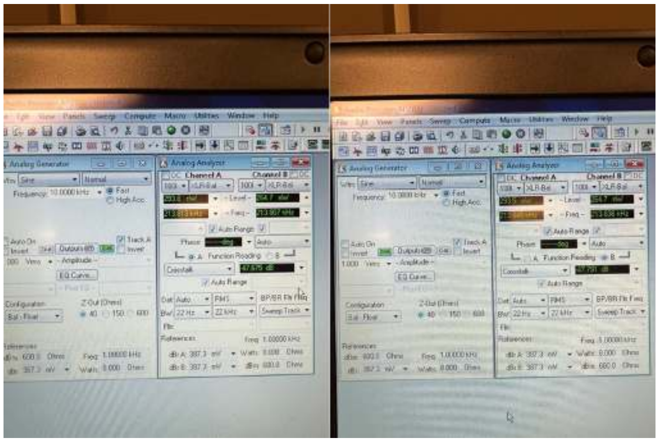
Channel A: 0.00108% THD+N, 1 kHz at rated power, 22kHz BW.
Channel B: 0.00113% THD+N, 1 kHz at rated power, 22kHz BW.

Phase:



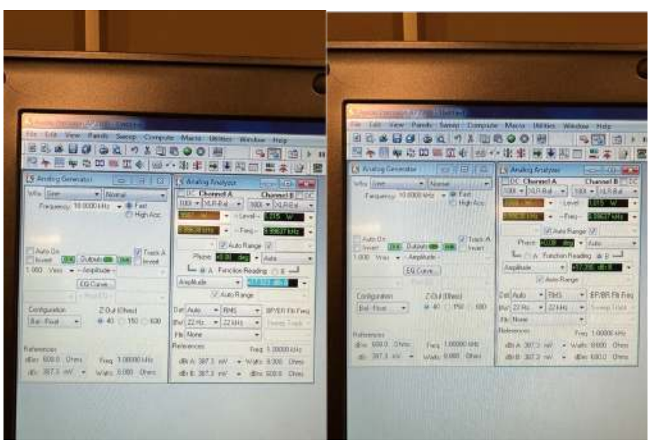
The charts above show the results for phase at frequencies of 1,000, 100, 10 kHz. Phase measurements describe the positive/negative time offset of waveforms in a cycle. Additionally, phase between channels can vary with frequency. The results differed slightly with readings of +0.02, +0.08, +0.12 degrees.

Crosstalk:



The charts above show the crosstalk between channels A and B. To setup this measurement the Analog Generator was set to an amplitude with the frequency at 10 kHz. Crosstalk was selected in the function reading menu to display signal leakage between channels A and B. The readings from our measurement were -47.675 dB in channel A and -47.791 dB in channel B. Our results differed from the manufacturers readings < -110dB, -115dB (typical) due to the parameter differences.

Signal to Noise ratio:



The charts above show the signal to noise ratio of both channels. To setup this measurement the analog generator was set to amplitude, then bandwidth fields were adjusted to 22 Hz and 22 kHz. To capture this reference the analyzer was set to dBr. Channel A had a reading of +17.323 dBr and channel B +17.390 dBr. The results of this measurement differed from the manufacturers noise readings of < -86dBu, -91dBu (typical) due to the parameters being different. Overall, if adjustments were made to match the manufacturers, I believe we would see similar results for all the measurements we conducted.

Works Cited

[1] SSL. (2018). Fusion User Guide [Online]. Available: https://www.solidstatelogic.com/assets/uploads/downloads/Fusion_User_Guide_V1.4.0.pdf. [Accessed 24 April 2023].